

Emulfree® CBG MB

Cooling sensation



People make our name



A unique structure called a bi-gel



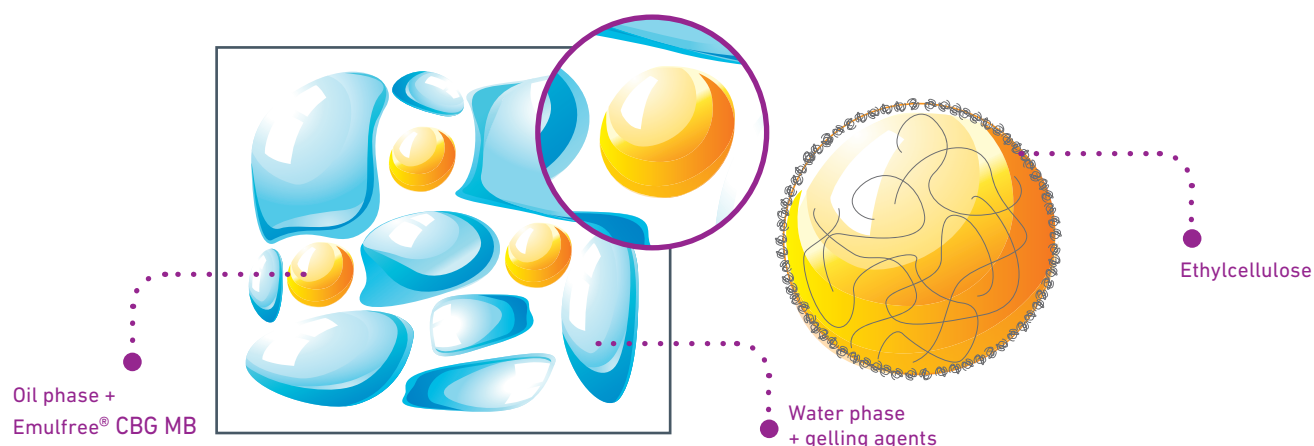
Textures with a watery sensation offer an immediate feeling of enhanced moisturization.

Emulfree® CBG MB is a unique ingredient that brings a refreshing, long-lasting sensation of coolness upon application.

To provide the skin with the right hydrating and nourishing levels, it is necessary to bring both hydrophilic and lipophilic components. This can generally be achieved using emulsions stabilized by surface-active molecules. However, the use of surfactants on sensitive skin can lead to irritation and dehydration.

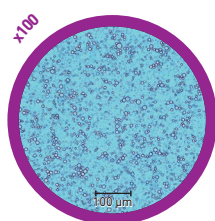
What is a bi-gel?

A bi-gel is a dispersion of an oily network within an aqueous gel, stabilized by a unique liquid surfactant-free system: Emulfree® CBG MB. This lipid-based system is a synergistic association of ethylcellulose with emollients that stabilizes the oil-water interface and allows the creation of refreshing, **quick-break** textures.

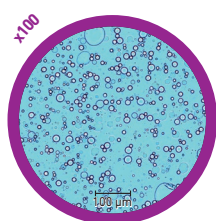


Ethylcellulose, located both in the oil droplets and at the oil-water interface, stabilizes the droplets via steric repulsion and prevents flocculation and coalescence.

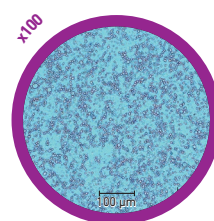
[A bi-gel presents perfect stability comparable to emulsions]



Bi-gel
with Emulfree® CBG MB



Dispersion
without Emulfree® CBG MB



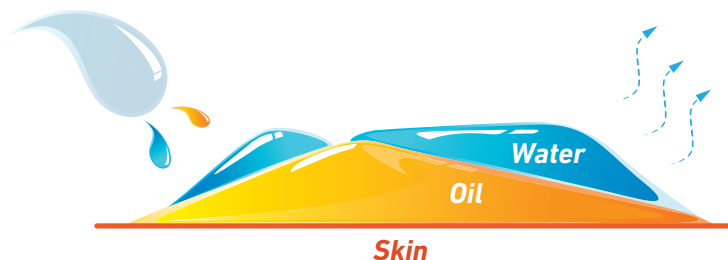
Emulsion
with a surfactant

Emulfree® CBG MB enables the association of an oil phase and a water phase without any surfactant. This unique structure, called a bi-gel, gives **innovative technical and sensorial properties** to cosmetic formulations.

Multi-benefits light and fresh textures

Upon application on the skin, the bi-gel breaks liberating the two phases: a fine oily layer stays on the skin while a fraction of the water evaporates.

This unique structure offers multiple benefits to cosmetic formulations.



Ideal for sensitive skin

The absence of potentially irritating surfactants makes the bi-gel ideal for sensitive skin. The composition of Emulfree® CBG MB has been carefully studied to **avoid any skin reaction** and **ensure perfect comfort**.



Key in sun protection

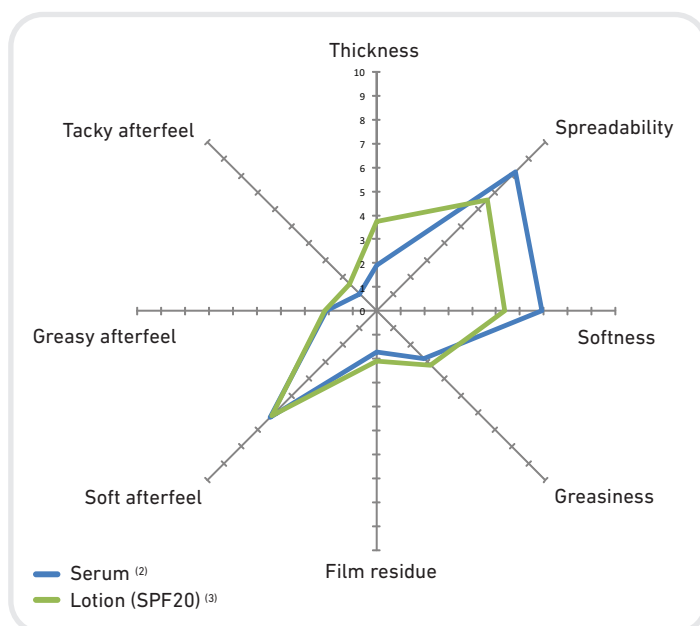
Emulfree® CBG MB naturally increases water-resistance of sunscreen formulations **without the use of water resistance agents**. A bi-gel containing 3% of hydrosoluble filters has shown 62% water resistance *in vivo* ⁽¹⁾.



Make-up removal efficacy

Upon application, the oil phase becomes available to **dissolve make-up** and **lift pigments off the skin**.

Waterproof black mascara
+ a bi-gel vs an O/W emulsion



Emulfree® CBG MB is ideal for the formulation of ultra light, fresh textures. Non greasy, non tacky and with a light residual film, it offers an easy and comfortable spread.

Easy to spread, Emulfree® CBG MB yields ultra-light textures that create a **cooling sensation**. Non greasy, non tacky, the residual film after complete absorption is almost imperceptible.

⁽¹⁾ Evaluation of water resistance after a single application and solar exposure. Tested on 5 Caucasian subjects, on the skin of the back, after 40 minutes of water immersion into flowing water at about 29 ± 2 °C according to water-resistance COLIPA guidelines.

⁽²⁾ 1693-1.02

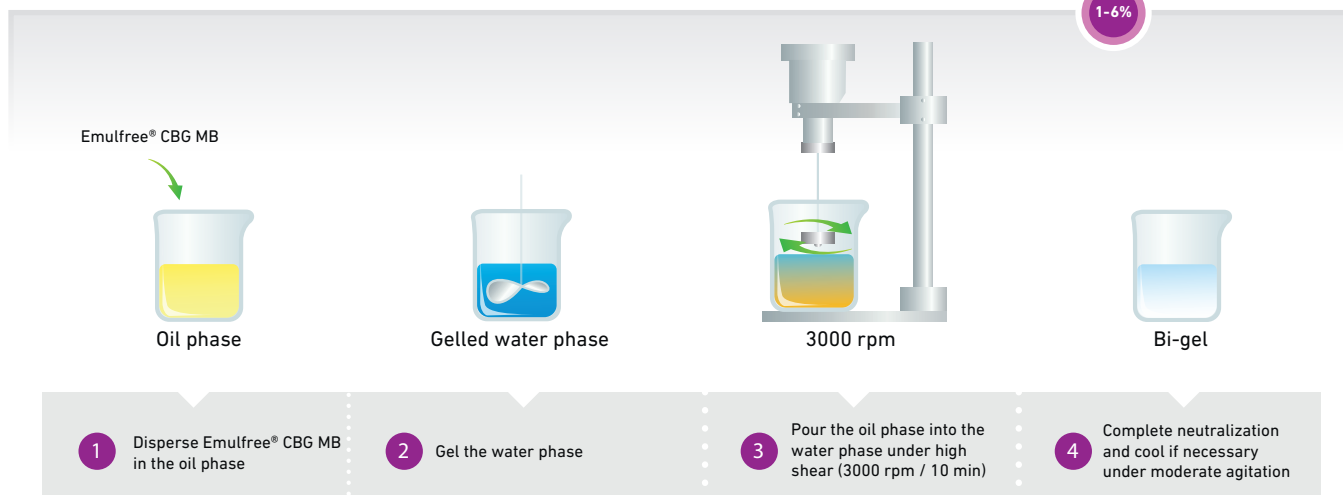
⁽³⁾ 1633-1.13

From sprays to creams

Process

Use level

1-6%



Emulfree® CBG MB is surfactant-free so high energy is necessary to create a perfect bi-gel. Hot process is recommended to ensure fine dispersion and stability but cold process is possible.

Compatibilities

Oils	Up to 30%
Actives	Compatible with most cosmetic active ingredients Issues are observed with resins (such as <i>VP/Eicosene Copolymer</i>) and certain cationics (<i>Steapyrium Chloride</i>)
Organic UV filters	Yes
Mineral UV filters	Low: up to 5% ZnO, TiO ₂
Alcohol	Up to 20%
Electrolytes	Low (depends on aqueous gelling system)
pH	3-10 (depends on aqueous gelling system)

Recommended associations

	Use Level			
	Spray	Lotion	Gel cream	Cream
Emulfree® CBG MB	1-4%	2-3%	4-6%	5-6%
Gelling agent	NECESSARY			
	Pemulen® TR2 or Xanthan Gum + Carbopol® Aqua SF1-0S	Xanthan Gum + Carbopol® ETD 2050	Xanthan Gum + Carbopol® ETD 2050 or Carbopol® Ultrez 10	Xanthan Gum + Carbopol® Ultrez 10 or Carbopol® Ultrez 21 or Novemer™ EC1 or Modified Potato Starch or Microcrystalline Cellulose Gum
	With acrylic polymers, partially neutralize before emulsification			
Consistency agent	No	No	No	Compritrol® 888 CG or Cetyl or Stearyl Alcohol

Emulfree® CBG MB



creates innovative textures with refreshing sensation and excellent ease of application. The absence of any surfactant enables to develop high tolerance formulations with unique properties.

- O/W stabilizer
- Strong sensorial identity
- Quick-break effect
- Watery sensation

Claims

- Surfactant free system
- Water resistant
- Ultra-light sensation
- Recommended for sensitive skin

Applications

- Skin care: all skin types
- Toiletries: after shave lotions, cleansers
- Suncare: waterproof ranges and sweatproof products
- Perfumed lines

Sensoriality

- Easy and comfortable spread
- Non greasy, non tacky after-feel
- Light residual film

Use level

1-6%

Use pH

3-10

As a user of palm oil and palm kernel oil derivatives, Gattefossé is sensitive to the environmental impact of palm tree growing and participates in the development of sustainable cultivation, providing its customers with environmentally responsible ingredients.

Gattefossé is a proactive member of RSPO (Roundtable on Sustainable Palm Oil) since 2010. In 2013, Gattefossé achieved the RSPO supply chain certification. Since then, Gattefossé offers products certified RSPO Segregated, and more recently, products certified RSPO Mass Balance. Emulfree® CBG MB is certified Mass Balance.

INCI: Isostearyl Alcohol (and) Butylene Glycol Cocoate (and) Ethylcellulose

Worldwide approval for cosmetic applications

Listed on IECIC (Inventory of Existing Cosmetic Ingredients in China) 2015

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